Design patterns and backend architecture

Chapter 1: How to study this guide

Practice designs, not claims about an employer.

1. How to study this guide

Design principle = a rule for making choices, such as keeping responsibilities clear. **Design pattern** = a reusable way to organize classes or objects for a recurring problem. **Architecture pattern** = a way to organize larger components and their communication. An implementation may use several patterns, but every pattern should solve a named problem.

In the upcoming technical round, prepare to explain a design choice while coding or discussing a project. For the later system-design round, practice drawing boundaries, data flow, failure paths, and tradeoffs. Do not spend this week memorizing a pattern catalog. Learn the problem, the mechanism, the cost, and when to avoid the pattern.

Fit into the existing 20-hour plan

During Java fundamentals — 35 minutes. Study SOLID, composition, immutability, Strategy, Factory, Builder. *Checkpoint:* Refactor one conditional-heavy example aloud.

During Spring and API study — 35 minutes. Study layering, Adapter, Repository, dependency inversion, transactions. *Checkpoint:* Trace a request and name each responsibility.

During project-story practice — 30 minutes. Connect one real decision to its tradeoff and test. *Checkpoint:* Give a two-minute design story with personal ownership.

During production review — 35 minutes. Study outbox, retry, circuit breaker, bulkhead, idempotency. *Checkpoint:* Explain timeout and duplicate-event behavior.

During a mock follow-up — 15 minutes. Practice selecting and critiquing patterns. *Checkpoint:* Answer without buzzwords.

That is **2 hours 30 minutes** inside your 20-hour plan. For the later system-design round, revisit the architecture section more deeply. If the interviewer asks about a technology you used, give its actual implementation details before naming a pattern.

Use this answer structure: **"The problem was __. I chose __ because __. The cost was __. I handled failure by __. We verified it with __."** Speaking the tradeoff is what makes an answer sound senior.